Sucheta Nandy

Philadelphia, PA | suchetanandy217@gmail.com | +1 267-235-0456
github.com/sucheta-nandy | sucheta-nandy.github.io | linkedin.com/in/suchetanandy

Professional Summary

Data and AI engineer with **3+ years of enterprise experience** building scalable pipelines, analytical systems, and data products. Graduate researcher applying **data engineering, machine learning, NLP, model evaluation, and product thinking** to reliable real-world systems.

Education

University of Pennsylvania *Aug 2025 - May 2027*
MSE in Data Science and Artificial Intelligence *Philadelphia, PA*
Maulana Abul Kalam Azad University of Technology *Jul 2018 - Jun 2022*
Bachelor of Technology in Computer Science and Engineering *India*

Technical Skills

Analytics & Modeling: Python, SQL, R, statistics, classification, cross-validation, feature engineering, model evaluation
Machine Learning & AI: PyTorch, TensorFlow, scikit-learn, XGBoost, LangGraph, RAG, Graph RAG, Neo4j
Data Engineering: PySpark, Databricks, Spark, BigQuery, Kafka, Airflow, ETL/ELT, data warehousing, DuckDB, Oracle
Product & Engineering: React, JavaScript, Tableau, Qlik, GCP, Azure, Docker, Git, CI/CD, Linux/Unix, automated testing
Certifications: Google Cloud Professional Data Engineer; Google Cloud Associate Cloud Engineer

Selected Data and AI Projects

ReviewIQ | Adaptive Travel Review Intelligence
github.com/sucheta-nandy/reviewiq

- Developed an adaptive review experience over **5,999 Expedia reviews across 13 properties**, detecting missing, sparse, and stale information across **16 travel topics**.
- Built a **multi-signal gap-scoring engine** and **contextual follow-up flow** that avoids topics a reviewer already covered and asks one or two targeted questions.
- Delivered a **React and Vite** application with text, quick-reply, and **Web Speech API** input plus animated completeness and impact visualizations.

Netflix Data Engineering Pipeline | PySpark ETL and Data Quality
github.com/sucheta-nandy/netflix-data-pipeline

- Engineered a **modular PySpark ETL workflow** covering ingestion, cleaning, enrichment, validation, and query-ready warehouse loading.
- Implemented **schema, completeness, uniqueness, range, and business-rule checks** before and after transformation, with automated JSON quality reports.
- Optimized analytical storage with **partitioned Parquet and Snappy compression** and added configuration, logging, unit tests, and integration tests.

Experience

Honeywell *May 2026 - Aug 2026*
Data Engineering Master's Intern *Atlanta, GA*

- Designed an enterprise analytical system using **Python, Databricks, Docker, LangGraph, Neo4j, and Graph RAG**, integrating retrieval, generation, and validation workflows that achieved **79% accuracy** and reduced manual query-development effort by **40%**.
- Engineered reusable data components connecting user questions with database schemas, relationships, metadata, and business rules for reliable analytics over complex enterprise datasets.
- Built automated evaluation, data-quality, validation, and monitoring workflows to identify incorrect outputs and systematically analyze model and pipeline failures.
- Developed a data and AI architecture combining knowledge graphs and semantic retrieval, contributing to an internal **IP/patent disclosure**.

Tata Consultancy Services *Aug 2022 - Aug 2025*
Data Engineer *India*

- Engineered enterprise data-processing systems using **Python, SQL, Oracle PL/SQL, BigQuery, and GCP**, improving performance by **25%** and enabling low-latency delivery.
- Automated ETL and deployment workflows using Unix shell scripting, Git, CI/CD, BigQuery, and GCP, reducing release cycles by **30%** across cloud and on-premises environments.
- Built automated validation and reconciliation workflows that contributed to a **98% defect-free release rate** and improved downstream data accuracy to **96%**.
- Developed Tableau and Qlik reporting outputs and collaborated with business stakeholders to translate data into operational insights.

University of Pennsylvania *Aug 2025 - Present*
Graduate Teaching Assistant *Philadelphia, PA*

- Support graduate coursework in **statistics, machine learning, and data analysis**, explaining quantitative concepts and reviewing analytical work.
- Re-engineered **3+ research workflows from R to Python** and automated batch processing and reporting, reducing recurring manual analysis effort by approximately **45%**.